

1. Administrative & Study Identification

Full Study Title: Long-term Follow-up (LTFU) of Patients Treated With Genome-edited Autologous Hematopoietic Stem and Progenitor Cells (HSPC) **Short Title/Acronym:** OTQ923 LTFU **Protocol Number:** CADPT03A12001 **NCT Number:** NCT06155500 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** OTQ923 (ex vivo CRISPR-Cas9 genome-edited autologous hematopoietic stem and progenitor cells targeting the γ -globin promoter) **Phase of Development:** Phase N/A (Observational) **Medical Monitor:** Novartis Clinical Operations Lead / Medical Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To monitor patients treated with OTQ923 for long-term safety and efficacy for a total of 15 years following infusion. **Secondary Objectives:** To assess the long-term durability of fetal hemoglobin (HbF) induction, prevention of severe vaso-occlusive crises (VOCs), overall survival, and to monitor for any delayed adverse events such as insertional oncogenesis or clonal expansion.

2.2 Background & Rationale Sickle Cell Disease (SCD) is a severe, inherited monogenic blood disorder caused by a mutation in the β -globin gene, leading to polymerized sickle hemoglobin (HbS), which results in red blood cell sickling, hemolytic anemia, and recurrent vaso-occlusive crises. OTQ923 is an investigational therapy that utilizes CRISPR-Cas9 gene editing to reactivate fetal hemoglobin (HbF) production, which prevents HbS polymerization. Because gene therapies involve permanent genomic modifications and conditioning regimens, regulatory agencies (e.g., FDA) mandate a 15-year long-term follow-up to ensure sustained efficacy and monitor for delayed safety events.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational **Control Type:** Single-arm (following all enrolled patients from the parent trial CADPT03A12101/NCT04443907) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N\$: Determined by the number of patients treated in the parent protocol (CADPT03A12101) **Number of Sites:** Multi-center (United States) **Estimated Study Start:** April 2024 **Estimated Primary Completion:** January 2039

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients must be 18 years and older. 2. Accepted invitation to join based on prior treatment with gene therapy. 3. Patients must have received gene therapy treatment (OTQ923) from the parent treatment protocol (CADPT03A12101). 4. Patients must provide informed consent prior to their entry into this study.

4.2 Exclusion Criteria 1. Completion of less than 1 year of safety follow-up in the parent treatment protocol (CADPT03A12101). 2. Withdrawal of consent or inability to comply with long-term follow-up requirements.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: N/A. No new investigational product is administered in this study. Patients were previously treated with a single infusion of OTQ923 following myeloablative conditioning in the parent trial. **Route of Administration:** N/A (previously Intravenous) **Duration of Treatment:** 15 years of observational follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care for SCD management as clinically indicated, as well as any supportive care post-transplantation. **Prohibited:** Further investigational gene therapies that could confound long-term safety and efficacy evaluations.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening / Rollover	Year 1 Post-Infusion
Informed Consent, Medical History, Baseline LTFU Labs		Interim	
Annual	Years 2-14	Safety Labs, Efficacy Assessment (HbF levels, total Hb, VOC tracking), Malignancy Screening	End of
Study	Year 15	Final Vital Signs, Exit Interview, Final Blood/Bone Marrow Assessments	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Long-term safety profile, specifically the incidence of delayed adverse events, gene-therapy-related toxicities (including insertional oncogenesis, hematologic malignancies, or clonal dominance), and overall survival over the 15-year follow-up period. **Measurement Method:** Clinical assessments, laboratory diagnostics, and specialized genetic/molecular monitoring of peripheral blood and bone marrow.

7.2 Secondary & Exploratory Endpoints - Long-term maintenance of HbF and total hemoglobin concentrations over time. - Proportion of participants with sustained resolution of severe Vaso-Occlusive Crises (VOCs). - Multi-lineage persistence of genome-edited HSPCs.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine annual reporting focusing on events of special interest (e.g., new malignancies, immunologic/hematologic disorders, and conditioning-related late effects). **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline, with immediate escalation for any suspected secondary malignancies. **Data Monitoring Committee (DMC):** External Data Monitoring Committee to review long-term safety trends and ensure patient safety across the gene therapy portfolio.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Full Analysis Set (FAS), consisting of all patients enrolled in the LTFU study who received an infusion of OTQ923. **Sample Size Justification:** N/A. The sample size is strictly dependent on the number of patients successfully treated in the CADPT03A12101 parent trial. **Handling of Missing Data:** Descriptive statistics will be utilized. Missing data will be handled based on available cases at each annual follow-up time point, without imputation for long-term safety events.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP), the Declaration of Helsinki, and FDA/EMA guidance on long-term follow-up for gene therapy products. **Monitoring Plan:** Periodic data collection, combined with remote and onsite monitoring visits corresponding with the annual follow-up milestones. **Audit Trail:** Robust Electronic Data Capture (EDC) systems equipped with complete audit trails to secure 15-year longitudinal patient data.

11. Study Identifiers & Governance

Primary ID: CADPT03A12001 **Registry Number:** NCT06155500 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Long-term Follow-up (LTFU) of Patients Treated With Genome-edited Autologous Hematopoietic Stem and Progenitor Cells (HSPC) **Official Regulatory Title:** Long-term Follow-up (LTFU) of Patients Treated With Genome-edited Autologous Hematopoietic Stem and Progenitor Cells (HSPC)

12. Clinical Framework (Sickle Cell Disease Specific)

Primary Condition: Sickle Cell Disease (SCD) **Diagnosis Threshold:** History of severe SCD, prior enrollment, and dosing in the OTQ923 parent trial **Medical Methodology:** Ex vivo CRISPR-Cas9 genome-edited autologous HSPCs **Optimization Target:** Sustained γ -globin (HbF) induction and VOC elimination

13. Study Procedures & Methodology

Management Pathway: Long-term post-gene therapy clinical monitoring pathway **Education Protocol (HCP):** LTFU safety reporting requirements, vigilance for delayed malignancy/genotoxicity **Education Protocol (Patient):** Importance of 15-year longitudinal follow-up, symptom reporting **Quality Audit Mechanism:** Centralized registry, genetic sequence persistence tracking, and annual clinical assessment completion rate

14. Observation & Follow-Up Schedule

Total Duration: 15 Years (180 months) post-infusion **Onsite Visit Cadence:** Annually (Years 2 through 15) **Remote Monitoring Frequency:** Periodic interim telehealth contacts based on clinical needs **Checklist Review Interval:** Annually

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				